Discovery of Linguistic Patterns in Japanese News Articles

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***Abstract*—This project applies unsupervised data mining techniques to a large corpus of Japanese news articles collected between 2005 and 2021. The goal is to discover latent structure in real world Japanese text without relying on predefined labels or prediction tasks. By using document clustering and association rule mining, this project aims to uncover emergent thematic groupings and recurring linguistic patterns that characterize Japanese news writing across different sources and time periods.**

I. Introduction

Large scale text collections such as news archives contain rich linguistic and semantic structure that is not always explicitly labeled. In the context of Japanese-language data, this structure is further complicated by the lack of whitespace between words, frequent use of kanji compounds, and stylistic variation across news domains. This project focuses on discovering patterns that naturally emerge from the data itself.

There are two main motivations for doing this project. First, it provides an opportunity to apply data mining techniques to non-English text, which introduces unique preprocessing and representation challenges. Second, the use of Japanese news articles aligns with my interest in the Japanese language, allowing discovered patterns to be interpreted both computationally and linguistically.

II. Dataset Description

A. Dataset Source

The dataset used in this project is the *Japanese Newspapers 2005–2021* dataset, publicly available on Kaggle [1]. It contains Japanese-language news articles collected from multiple online newspaper sources, including but not limited to (nikkei.com, tomamin.co.jp, and hokkaido-np.co.jp), over a sixteen-year period.

B. Dataset Structure

The dataset is provided in CSV format, where each row corresponds to a single news article. The primary attributes used in this project are summarized in Table I.

The dataset contains tens of thousands of articles and several million characters of text. Article lengths vary widely, ranging from short announcements to full-length reports.

Field	Description
source	Newspaper website or domain name
date	Article publication date
text	Full Japanese article content

TABLE I
Dataset schema

C. Data Characteristics and Quality Considerations

The dataset does not include topic labels or metadata describing article categories, making it particularly suitable for unsupervised discovery tasks. However, several preprocessing challenges must be addressed. Japanese text does not use whitespace to separate words, requiring morphological analysis to identify meaningful tokens. Additionally, articles may contain punctuation, numbers, or stylistic markers that do not contribute to semantic analysis.

Despite these challenges, the dataset represents authentic, real-world language use across a wide range of topics and writing styles. Each article is treated as an independent document for analysis purposes.

III. Discovery Questions

This project investigates the following discovery-oriented research questions.

A. Q1: Document Clustering

Are there natural clusters of Japanese news articles based on linguistic similarity, and what types of themes or writing styles emerge within these clusters?

This question explores whether articles naturally group together based on vocabulary usage and linguistic structure, without relying on predefined categories. By clustering documents using unsupervised techniques, the project seeks to uncover latent groupings that may correspond to broad themes such as politics, economics, culture, or local events, as well as stylistic differences between sources.

The value of this question lies in understanding how thematic structure emerges organically from language usage, rather than being imposed by human labeling.

B. Q2: Linguistic Associations

Which Japanese words or phrases frequently co-occur within news articles, and what semantic patterns do these associations reveal?

This question focuses on identifying frequently co occurring words or phrases that reflect common expressions, domain specific terminology, or recurring narrative patterns in Japanese news writing. Discovering such associations can reveal how concepts are linguistically linked in practice, offering insight into both journalistic conventions and real world language use.

IV. Planned Techniques

A. Text Preprocessing

Prior to analysis, the text data will undergo several preprocessing steps. These include morphological tokenization using a Japanese tokenizer, removal of stopwords and punctuation, and normalization of text where appropriate. Each article will then be represented in a numerical format suitable for data mining techniques.

B. Clustering Methods

To address Q1, clustering techniques such as K-Means or Hierarchical Clustering will be applied to document representations such as term-frequency or TF-IDF vectors. The resulting clusters will be analyzed by examining representative documents, frequently occurring terms, and source distributions within each cluster. This analysis will focus on interpretation rather than cluster accuracy.

C. Association Rule Mining

To address Q2, association rule mining techniques such as Apriori or FP-Growth will be applied. Each document will be treated as a transaction containing a set of words. Frequent itemsets and strong association rules will be analyzed to identify meaningful word relationships and recurring linguistic patterns.

D. Interpretation and Validation

Because this project emphasizes discovery, results will be evaluated qualitatively. Cluster coherence and association rules will be interpreted through inspection of example articles and linguistic analysis, rather than predictive metrics.

V. Data Loading and Overview

We use the *Japanese Newspapers 2005 - 2021* dataset from Kaggle. Each row is a Japanese news article with the following fields:

- **source**: newspaper domain (e.g., `nikkei.com`)
- **date**: publication date
- **text**: full Japanese article

After dropping rows with missing or empty `text`, we obtain **312,954** articles. Article length in characters is quite varied: the median is about 100 characters, with many short and medium-length pieces and a smaller number of very long articles (up to about 5,000 characters).

VI. Preprocessing and Tokenization

Because Japanese does not use spaces between words, we apply morphological analysis with the `fugashi` library. For each article we obtain a list of tokens tagged with part of speech.

To make the representation more useful for mining, we then:

- Keep only nouns (名詞), verbs (動詞), and adjectives (形容詞).
- Remove tokens that are only numbers (digits or full-width digits).
- Remove very common, low information tokens such as:
  - light verbs and generic nouns: する, いる, ある, こと, よう, etc.
  - time-related characters: 年, 月, 日
  - single kana characters and common punctuation like 。, ー, ・, %.

After this step, each article is represented as a cleaned list of meaningful Japanese word tokens (for example, `[" 震災", " 被災", " 支援"]`). We also record simple features like the number of characters and the number of tokens per article.

VII. Exploratory Data Analysis

A. Length and Source Distributions

We analyze article lengths in terms of both characters and tokens. Histograms of the full dataset show a right-skewed distribution, with most articles being short to medium in length and a small number of very long pieces, which is typical for news content.

To reduce the influence of extreme values, we focus on the 95th percentile of lengths. Within this range, the distributions of both character and token lengths appear approximately normal, indicating that the bulk of the dataset is symmetrically distributed around the mean.

We also plot a bar chart of the number of articles per `source`. A few major outlets contribute a large portion of the data, while many smaller outlets contribute fewer articles.

B. Most Frequent Tokens

Using the preprocessed tokens, we examine the most frequent words overall and for a few major sources. Common tokens include:

- Places and general nouns such as 市 (city), 県 (prefecture), 町 (city/town), 日本 (Japan).
- Economic terms such as 円 (Yen), 万 (Ten thousand), 億 (One hundred million).
- Disaster-related words such as 震災 (Disaster), 被災 (Disaster damage), 原発 (Nuclear power plant).
- Sports terms such as 選手 (Pro athlete), 大会 (Convention) , 試合 (Match).

Per-source frequency profiles reflect the domain of each outlet. For example, `nikkei.com` contains many economic

terms (投資, 企業, 市場, 株式), while some regional papers have many local place names and disaster-related vocabulary.

VIII. K-Means Clustering

A. TF–IDF Representation

To use the articles in a clustering algorithm, we convert each tokenized article into a numeric vector. We join tokens with spaces (so that each Japanese word remains a single token) and use scikit learn's `TfidfVectorizer` to build a sparse document - term matrix.

We ignore very rare words, and drop words that appear in more than half of the documents. The result is a large, sparse matrix X where each row corresponds to an article and each column corresponds to a Japanese token. The entries are TF–IDF weights that indicate how characteristic each token is for each article.

1) Feature Scaling Considerations: Distance based clustering algorithms are sensitive to feature scale. However, TF - IDF vectors already represent normalized term weights that reflect the relative importance of each token within a document compared to the overall corpus. Because of this normalization, additional standardization was not necessary before applying K-Means. Using TF - IDF directly allows the distance metric to capture differences in term importance between articles while preventing very frequent tokens from dominating the similarity calculation.

B. Clustering Method

To address Q1 ("Are there natural clusters of Japanese news articles and what themes emerge?"), we apply **K-Means** clustering to the TF - IDF vectors.

We use the following settings:

- Number of clusters: $K = 8$.
- Implementation: scikit-learn `KMeans` with a fixed random seed and multiple restarts.

Each article in the sample receives a cluster label from 0 to 7. We then:

- Count how many articles fall into each cluster (and plot a simple bar chart).
- Look at the top TF - IDF tokens for each cluster (the most important words).
- Check which news sources appear most often in each cluster.

1) Choosing the Number of Clusters: Selecting an appropriate value for K is an important step when applying K-Means clustering. We evaluated several possible values of K using the elbow method. As expected, the WCSS decreases as K grows, but the rate of improvement slows after a certain point.

As shown in Figure 1, the elbow plot suggests diminishing returns beyond roughly 6 - 9 clusters. We chose $K = 8$ as a compromise. Using more clusters produced only small improvements in the objective function while making the resulting clusters harder to interpret.

C. Cluster Themes and Findings

The clusters produced by K-Means line up well with intuitive news themes:

- **Cluster 0: National and political news.**
  Contains words such as 日本 (Japan), 首相 (Prime Minister), 原発 (Nuclear power plant), 政府 (Government), 問題 (Issue), 中国 (China) which suggest national policy, major events, and international relations.
- **Cluster 1: Finance and markets.**
  Dominated by 円 (Yen), 万 (Ten thousand), 億 (Hundred million), ドル (Dollar), 市場 (Market). This cluster focuses on money, prices, and markets, and includes many articles from `nikkei.com`.
- **Cluster 2: Local and regional news, including disaster coverage.**
  Important tokens are 市 (City), 県 (Prefecture), 町 (Town), 区 (Ward), 市民 (Citizen), 地域 (Region), 福島 (Fukushima), 震災 (Earthquake disaster). This cluster collects municipal and prefectural stories, especially about the Tohoku earthquake and its aftermath.
- **Cluster 3: Event and schedule information.**
  Tokens like 時 (Hour), 午後 (Afternoon), 午前 (Morning), 分 (Minute), 無料 (Free), indicate event listings and notices with dates, times, and contact details.
- **Cluster 4: Crime and police reports.**
  Contains 容疑 (Suspect), 逮捕 (Arrest), 疑い (Suspicion), 捜査 (Investigation), 事件 (Incident). Articles in this cluster are typical crime stories about incidents, arrests, and investigations.
- **Cluster 5: Baseball game reports.**
  Includes 回 (Inning), 点 (Point / Run), 安打 (Hit), 失点 (Runs allowed), 試合 (Game / Match), 大会 (Tournament), 投手 (Pitcher), 野球 (Baseball), 先発 (Starting player), 三振 (Strikeout). These are detailed reports on baseball games at both professional and high school levels.
- **Cluster 6: Opinion and feature-style writing.**
  Uses more subjective and reflective language such as 自分 (Oneself), 考え (Thought), 思い (Feeling), 今 (Now), 世界 (World), 言葉 (Words), suggesting columns, interviews, and human-interest features.
- **Cluster 7: Other sports and major competitions.**
  Tokens like 選手 (Player / Athlete), 大会 (Tournament), 決勝 (Final), 出場 (Participation), 優勝 (Championship / Victory), チーム (Team), 監督 (Coach), リーグ (League),indicate non-baseball sports, including the Olympics and international tournaments.

Overall, these results show that unsupervised clustering on Japanese news articles, using only tokenized text and TF IDF features, can recover clear and meaningful themes without any manual labels. This provides a positive answer

to the first research question in the project.

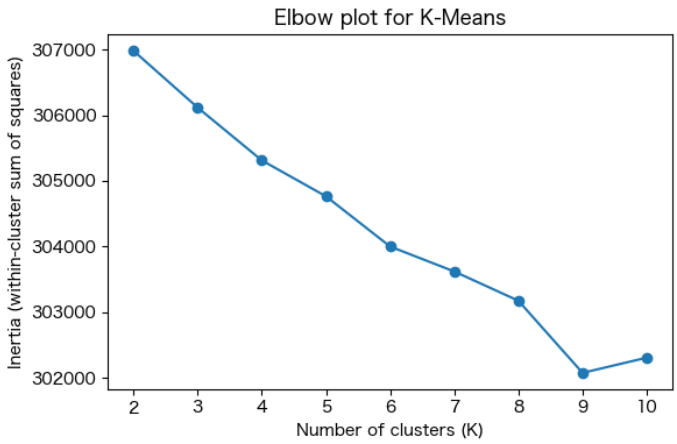

Fig. 1. Elbow method for selecting the optimal number of clusters

IX. FP-Growth

A. Analytical Choices

1) Transaction Representation: For association rule mining, every article becomes a *transaction*: the set of distinct content tokens that survive the preprocessing pipeline. Using sets (rather than bags) means repeated tokens in the same article count only once, which is the standard market-basket assumption.

2) Vocabulary Filtering: With over 312,000 articles the raw vocabulary is very large. To keep the one-hot transaction matrix tractable we require every token to appear in at least MIN_DOC_FREQ = 300 articles. This retains roughly 3,500 tokens that are common enough to participate in meaningful rules while discarding the long tail of rare terms that would inflate memory and runtime without producing interpretable patterns.

Transactions with fewer than two qualifying tokens are dropped, leaving approximately 307,000 usable transactions.

3) Algorithm: FP-Growth over Apriori: The M1 proposal mentioned Apriori as a candidate algorithm. During implementation we found that Apriori's candidate-generation step is prohibitively expensive on a vocabulary of several thousand items (the number of candidate pairs alone exceeds tens of millions). We therefore switched to **FP-Growth**, which builds a compressed prefix tree and mines frequent itemsets directly without generating candidates. FP-Growth produces identical itemsets and rules but completes in minutes rather than hours on this corpus.

4) Rule Parameters: We use `min_support = 0.008` (a pattern must appear in at least ≈2,500 of 312,000 articles), min_lift = 1.25, and restrict antecedent size to at most 2 items with a single-item consequent for interpretability.

B. Results and Interpretation

FP-Growth produces a large set of frequent itemsets, a subset of which is shown in Figure 2, from which we derive association rules sorted by lift. The top 18 rules by lift reveal five clearly interpretable semantic clusters.

1) Crime Reporting (lift ≈ 44):

逮捕 (arrest) ⇒ 容疑 (suspect/suspicion) confidence 0.78, lift 43.8

This is the strongest rule. When 逮捕 appears in an article, 容疑 co-occurs 78% of the time — roughly 44 times more often than chance. The reverse direction has a confidence of 0.53 (many articles discuss suspects without mentioning an arrest). These terms define the *crime and police* language register of the corpus.

2) Mobile Phone (lift ≈ 43):

携帯 (mobile/portable) ⇒ 電話 (phone) confidence 0.76, lift 42.5

携帯電話 (mobile phone) is a standard compound in Japanese that is typically tokenized into two words by the morphological analyzer. The extremely high lift confirms that these two tokens are almost never used independently — their co-occurrence reflects a fixed lexical unit rather than a topical pattern.

3) Press Conferences (lift ≈ 36):

記者 (reporter) ⇒ 会見 (press conference) confidence 0.64, lift 35.7

記者会見 (press conference) is another compound that splits during tokenization. Its high lift means that when reporters are mentioned, it is overwhelmingly in the context of a press conference, reflecting a dominant narrative pattern in political and institutional reporting.

4) Fukushima Nuclear Accident (lift 18–34): Three tokens — 福島 (Fukushima), 原発 (nuclear plant), and 事故 (accident) — form a tightly interconnected triangle of pairwise and three-way rules:

- {福島, 事故} ⇒ 原発: confidence 0.92, lift 34.1
- {福島, 原発} ⇒ 事故: confidence 0.74, lift 30.2
- {事故, 原発} ⇒ 福島: confidence 0.69, lift 26.5
- 事故 ⇒ 原発: confidence 0.56, lift 21.0
- 福島 ⇒ 原発: confidence 0.50, lift 18.5

The strongest direction is toward 原発: when both 福島 and 事故 appear, the article discusses a nuclear plant 92% of the time. This cluster reflects the lasting impact of the 2011 Fukushima Daiichi nuclear disaster on the Japanese news landscape.

5) Tohoku Earthquake and Disaster (lift 19–24): A second, partially overlapping disaster cluster involves 震災 (earthquake disaster), 東 (east), 被災 (disaster-affected), and 日本 (Japan):

- {日本, 震災} ⇒ 東: confidence 0.94, lift 23.8
- {被災, 東} ⇒ 震災: confidence 0.98, lift 23.7
- {東, 日本} ⇒ 震災: confidence 0.88, lift 21.3

These tokens reconstruct the phrase 東日本大震災 (Great East Japan Earthquake), which is split across multiple tokens by the morphological analyzer. The very high

confidence (>0.88) in all directions shows that these words rarely appear in this combination outside of earthquake coverage.

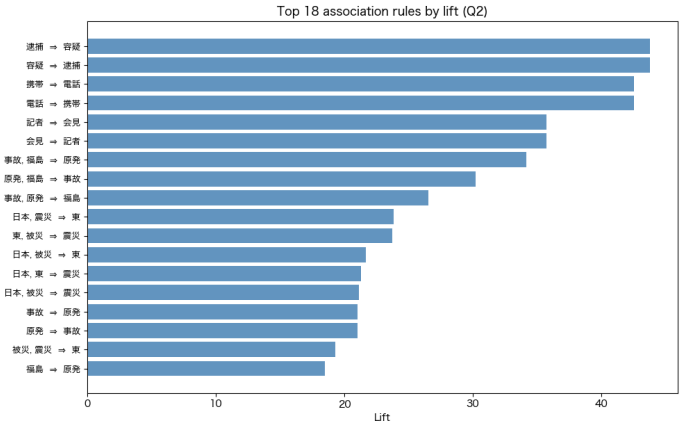

Fig. 2. Subset of association rules by lift

X. Conclusion

This project demonstrated that meaningful linguistic and thematic structure can be discovered from a large corpus of Japanese news articles using unsupervised data mining techniques. Without relying on predefined labels or supervised models, both K-Means clustering and FP-Growth were able to extract interpretable patterns from raw text data.

The K-Means clustering results showed that documents naturally group into coherent thematic categories, including politics, finance, regional news, crime reporting, and sports. These clusters emerged solely from TF-IDF representations of tokenized Japanese text, indicating that even simple vector-space models can capture high-level semantic organization in news corpora.

Association rule mining using FP-Growth further revealed strong co-occurrence relationships between Japanese terms. Many of the highest-lift rules corresponded to meaningful linguistic compounds or recurring journalistic expressions, such as crime-related terminology, disaster reporting language, and fixed lexical units that were split during morphological tokenization. In particular, disaster-related terms associated with the Fukushima nuclear accident and the Great East Japan Earthquake formed highly consistent and interpretable rule structures.

Overall, the results suggest that unsupervised techniques are effective for exploring structure in large-scale, unlabeled Japanese text data. While the findings are largely qualitative, they provide strong evidence that both topical clustering and lexical association analysis can uncover meaningful patterns in real-world news archives.

XI. Future Work

While the results of this project demonstrate that meaningful structure can be extracted from Japanese news articles using relatively simple unsupervised techniques, there are several directions for improving and extending this work.

One important improvement would be the use of more advanced text representations. TF-IDF does not capture word order or contextual meaning, which is especially limiting in Japanese due to compound expressions and flexible phrasing. Future work could explore neural embedding methods such as Word2Vec, FastText, or transformer-based models like BERT trained on Japanese corpora. These approaches may produce more semantically meaningful document representations and improve clustering quality.

For clustering, alternative algorithms such as hierarchical clustering or density-based methods like DBSCAN could be evaluated. These methods may better capture irregular cluster shapes and provide more insight into subtopic structures within larger thematic groups. Additionally, topic modeling techniques such as Latent Dirichlet Allocation (LDA) or BERTopic could be used as a complementary approach to compare against K-Means results.

In the association rule mining component, future work could incorporate phrase-level tokenization to reduce issues caused by morphological splitting of compound words. This would help distinguish true semantic relationships from artifacts of tokenization. More advanced sequential pattern mining methods could also be explored to capture ordering information between terms.

Another promising extension is temporal analysis of the dataset. Since the corpus spans from 2005 to 2021, analyzing how cluster distributions and word associations evolve over time could provide insight into changing news trends, such as shifts in political discourse, disaster reporting, or economic terminology.

Finally, integrating source-aware or domain-aware modeling could further improve interpretability. Different newspapers exhibit distinct writing styles and topical biases, and explicitly modeling these differences may lead to more refined and meaningful clusters.

XII. GitHub Repository

A GitHub repository has been created to manage the project code, documentation, and data organization throughout the semester [2].

The repository includes a `README.md` file that provides an overview of the project, including the project title, a brief description of the research goals, the dataset source link, and author information. This ensures the project is clearly documented and reproducible.

The repository is organized using the following structure:

- `data/` - Contains the dataset or placeholders with a
- `notebooks/` - Will contain Jupyter notebooks used for preprocessing, clustering, and association rule mining in later milestones.

- `docs/` - Will contain all documents related to this research, such as this proposal.
- `README.md` - Provides a high-level description of the project and dataset.

References

[1] V. Huholl, "Japanese Newspapers 2005–2021," Kaggle, 2021. [Online]. Available: https://www.kaggle.com/datasets/vyhuholl/japanese-newspapers-20052021

[2] W. Bramblett, "Discovery of Linguistic Patterns in Japanese News Articles (Code Repository)," GitHub, 2026. [Online]. Available: https://github.com/WyattBram/CS4412-DataMiningProject